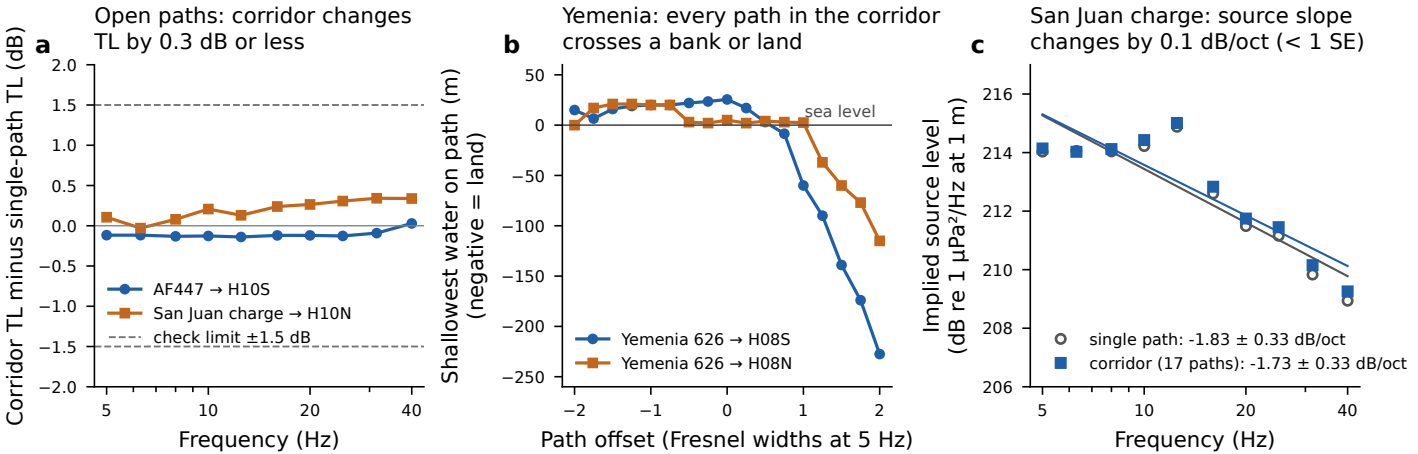
Sound loss calculated along a corridor of paths for the calibration events



Plain-language note: We calculate the sound loss along 17 paths side by side (a corridor) instead of along one line. (a) Where no seabed feature is in the path, the corridor gives almost the same loss as one path, so the method agrees with the earlier one. (b) From the 2009 crash site off the Comoros to the Diego Garcia hydrophones, each path crosses water shallower than 26 m or land, so the model sends no sound through. (c) For the 2017 calibration charge off Argentina, the corridor changes the calculated source spectrum only a little.

Technical note: N×2D Fresnel corridor (pre-registered nx2d.py 4d62d144): members offset u·F(s), F = sqrt(λ s (R−s)/R), u = −2..2 step 0.25, weight exp(−u²), KRAKEN coupled-mode per member on GEBCO_2026 (3×3 median, 0.5 km steps), hard bottom, receiver at hydrophone depth; a member with no propagating mode on any profile contributes zero. Checks on AF447 → H10S (z_s 10 m): K1 |corridor − ocean-transport section| ≤ 0.14 dB, K2 |u = 0 − section| ≤ 0.16 dB, both PASS (limits 1.5 and 1.0 dB). (a) z_s = 10 m; San Juan corridor − section -0.03 to +0.34 dB. (b) all 510 members per station blocked at 5–40 Hz (z_s 2, 10, 30 m): Saya de Malha Bank (9.2–9.9° S, 59.7–61.0° E), Farquhar group (10.2° S, 51.0° E) on H08N, Grande Comore on u > 0. (c) z_s = 33 m (charge depth, Vergoz 2021); received levels from Nielsen 2021 Fig. 5c (H10N), noise-subtracted; OLS on log2 f, 10 bands; absolute source level untested (no cited explosive spectrum yet). SOFAR branch only; 3-D diffraction beyond ±2 Fresnel widths and seabed-borne paths not modelled. Not used for any likelihood.